THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET B

Chapter 9: **Light – Reflection and Refraction** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The focal length of a plane mirror is: 1

(a) 1 m **(b)** infinite **(c)** negative **(d)** zero

2. The power of a lens is −2.5 D. It is a: 1

(a) concave lens of focal length 40 cm **(b)** concave lens of 25 cm
(c) convex lens of 2.5 m **(d)** convex lens of focal length 40 cm

3. The magnification produced by a plane mirror is: 1

(a) −1 **(b)** 0 **(c)** +1 **(d)** infinite

4. A student states: “The refractive index of diamond (2.42) means light travels fastest in it.” The correct response is: 1

(a) Correct
(b) Correct only for red light
(c) Incorrect — it is optically the densest; light is slowest
(d) Cannot be determined

5. For an object at infinity in front of a concave mirror, Student A says the image is at F; Student B says it is at C. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

6. The mirror used by a dentist is: 1

(a) plane **(b)** concave **(c)** convex **(d)** cylindrical

7. A concave lens always forms an image that is: 1

(a) magnified **(b)** real and inverted
(c) at infinity **(d)** virtual, erect and diminished

8. The magnification of a convex mirror is always: 1

(a) zero **(b)** positive and less than 1
(c) greater than 1 **(d)** negative

9. The refractive index of water is 1.33. The speed of light in water is about: 1

(a) 4×10^8 m/s **(b)** 1.33×10^8 m/s **(c)** 3×10^8 m/s **(d)** 2.25×10^8 m/s

10. A ray passing through the principal focus of a concave mirror after reflection: 1

(a) passes through F again **(b)** retraces its path
(c) becomes parallel to the principal axis **(d)** passes through C

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. What is the radius of curvature of a spherical mirror? 2

12. Define 1 dioptre. 2

13. Why does a ray of light bend when it travels from one medium to another? 2

14. A student says that a convex mirror can form a magnified image. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. An object is placed 10 cm in front of a convex mirror of focal length 15 cm. Find the position and magnification of the image. 3

16. An object is placed 30 cm from a concave lens of focal length 15 cm. Find the image position and magnification. 3

17. Draw ray diagrams (describe) for image formation by a concave lens when the object is 3
(i) at infinity
(ii) between infinity and optical centre.

OR

Give three uses of convex mirrors and convex lenses (any three total). 3

18. The refractive indices of kerosene, turpentine and water are 1.44, 1.47 and 1.33. In which does light travel fastest? Which is optically densest? 3

OR

What is the relation between focal length and radius of curvature? Find R for a mirror with $f = -12$ cm. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Explain image formation by a concave mirror for object at 5
(a) beyond C
(b) at F
(c) between P and F, with nature of image.
(ii) Which position is used in a shaving mirror?

OR

(i) An object 5 cm high is placed 25 cm in front of a concave mirror of focal length 10 cm (describe the diagram).
(ii) Find the image distance and height.
(iii) State its nature. 5

20. **Case study:** A doctor prescribes a lens of power +2.0 D. 5
(i) Is the lens converging or diverging?
(ii) Find its focal length.
(iii) If an object is placed 25 cm from this lens, find the image distance.
(iv) Nature of the image?

OR

(i) What is refraction? State its laws.
(ii) What is the absolute refractive index?
(iii) Explain why a coin in a bucket of water appears raised. 5

END OF QUESTION PAPER — Total: 40 Marks